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Low-cost emerging technologies as a tool to support informal environmental education in children from vulnerable public schools of southern Chile

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ABSTRACT

Reform to the Chilean educational system seeks to improve public education, adapting the school curriculum to new technologies and the information revolution. Our study focused on highlighting potential issues with environmental and technological education, offering opportunities to complement the current Chilean curriculum with an interdisciplinary approach. During a five-week period interspersed through the school semester, 115 underprivileged children in seventh grade from five public schools of southern Chile participated in workshops based on Arduino technology to address current environmental problems caused by global change and anthropization. Surveys were conducted to participating students and included both open and closed-ended questions, which measured attitudes and perceptions towards science and the use of technology in classrooms and daily lives. Surveys were analysed quantitatively with basic statistics and qualitatively by building semantic networks of the relationships between different concepts expressed by the students. Results did not show significant changes in attitudes towards science or technology before and after the workshops but students reported mixed feelings about science. Indifference and fear emerged, coupled with the need for more interactive experiences. Curiosity and enthusiasm also surfaced when students were using new technologies. Since the school curriculum is generally focused exclusively towards the completion of specific curricular aims, highly vulnerable schools in Chile have few opportunities to access interactive, hands-on activities. Low-cost, emergent technologies, such Arduinos, have a high potential for improving student's attitude towards science and technology and their use should be considered when tailoring school curriculum in Chile and elsewhere.

Abbreviations: 21CC: twenty-first-century competencies; CONICYT: Comisión Nacional de Investigación Científica y Tecnológica

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(Chilean National Commission for Scientific and Technological Research); ED TECH: Technological Education; EE: Environmental Education; ITCs: Information and Communication Technologies; JUNAEB: Junta Nacional de Auxilio Escolar y Becas - National Board of School Aid and Scholarships; PjBL: Project Based Learning; SIMCE: Sistema de Medición de la Calidad de la Educación - Education Quality Measurement System; VI: Vulnerability Index; UACh: Universidad Austral de Chile

Introduction

The current educational system in Chile is characterised by inequality and spatial segregation, due to the social, economic and cultural characteristics of the country (Bellei, 2013; Valenzuela, 2008; Valenzuela, Bellei, & De Los Ríos, 2010). A large gap exists between private and public educational institutions (Drago & Paredes, 2011), where economically privileged children obtain significantly higher results in standardised Chilean tests[1] in both natural sciences and mathematics (MINEDUC, 2018; Valenzuela, Labarrera, & Rodríguez, 2008), compared to vulnerable students which often do not have access to an educational system with the minimum resources necessary for proper functioning (e.g. economic and human resources, infrastructure, pedagogical material) (Bellei, 2013; Drago & Paredes, 2011; Valenzuela et al., 2008). Also, while in public schools, efforts and resources are exclusively aimed at reviewing specific content and at improving performance on standardised tests, in elite schools, some efforts are also directed at incorporating and developing critical thinking and problem solving skills as the basis of a twenty-first-century world-class education (Reimers & Chung, 2019).

During the last two decades, Chile has sought to improve the quality and equity of education through the implementation of multiple reforms (Contreras & Elacqua, 2005). One example is the 1997 Chilean educational reform, which included digital literacy and information and communication technologies (ITCs) as fundamental transversal objectives for primary and secondary education. While these skills are important twenty-first-century competencies (21CC) for everyday life, work and well-informed citizenship, their implementation in the Chilean school curriculum has been insufficient (Bellei & Morawietz, 2016). As a second example, and among the most recent educational reforms, there is the 'Ley de inclusión escolar' (School Inclusion Law) (Biblioteca del Congreso Nacional, 2015), which generated the definitive dissociation between public and private education with the elimination of subsidised private schools by the end of 2017. This government incentive consisted of a 'voucher' for each student enrolled and complemented with a co-payment from each student's family. The main goal of the removal of subsidised private schools was to break down the barriers that have historically contributed to inequality in the country, such as profit, selection and co-payment, and sought to guarantee access to a good education independently from the families' ability to pay (MINEDUC, 2017). The School Inclusion Law is in the process of being implemented (MINEDUC, 2017), and although some progress has been made in integrating the new policies, educational inequality in Chile remains high (OCDE, 2018).

An important challenge in the Chilean educational system is the adaptation of the school curriculum to todaýs society, characterised by constant change, technologies and the information revolution (Cox, 2011). To acknowledge this fast speed worldwide demand, first in the United States, and slowly into other countries, the school curriculum has evolved into STEM Education. STEM is based on the idea of educating students in four specific disciplines – science, technology, engineering and mathematics – following an interdisciplinary and applied approach. Rather than teaching the four disciplines as separate and discrete subjects, STEM integrates them into a cohesive learning paradigm based on real-world application and project-based learning (PjBL). This kind of learning is active because the contents are applied in practice. It has been observed that students generate a greater commitment to each task and develop attitudes and abilities of high cognitive level (Thomas, 2000). During this project, we focused our workshops on two specific curricular disciplines environmental education (EE) and technological education (ED TECH), as both mark two key examples to investigate some of the deficiencies in the public Chilean educational system, and offer opportunities to improve, complement and expand the school curriculum towards STEM curricular education. In general, the main objective of EE is to generate environmental awareness in the current context of climate and global change (Programa de las Naciones Unidas para el Medio Ambiente, 1975). In the Chilean curriculum, EE is considered a transversal pedagogical skill, meaning that it is not taught as a subject itself but connected to other disciplines of primary and secondary education based at the discretion of each teacher (Cabezas, 1997; Torres, Benavides, Latoja, & Novoa, 2017). On the other hand, ED TECH in Chile is a discipline that mostly conceives informatics as a tool for solving problems of personal organisation (i.e. end-user software applications such as word processors and presentation tools) to the detriment of the development of skills that allow to respond to broader problems and in tune with current social challenges (i.e. computer programming) (González Campos, Olarte Dussán, & Corredor Aristizabal, 2017; Jones, Buntting, & De Vries, 2013).

The National Commission for Scientific Research and Technology (CONICYT Chile), as the entity in charge of strengthening the country's scientific and technological base, has been offering, since 1995, educational funds through a programme called 'EXPLORA-CONICYT'. The aim of this programme is to develop a scientific culture between schoolchildren and teachers strengthening critical thinking, reflection and understanding of the environment through a learning model based on constructivism philosophy. Constructivism is a philosophy of learning based on exposure and demonstration, which emerged as an alternative to traditional teaching methods. This method requires a significant participation of the students in the teaching-learning process, and knowledge is acquired directly through experiences and the construction of a personal vision of the world (Flick, 1993). In this line, both hands-on science and project-based learning (PjBL) are strategies that promote learning through active participation and independence. In particular, hands-on activities require the active participation of students through the manipulation of educational materials and elements (Flick, 1993); while in PjBL students build knowledge through teamwork and problem solving (Krajcik, Czerniak, & Berger, 1999; Thomas, 2000). Some examples of educational projects, based on hands-on learning and PjBL models, have demonstrated effectiveness gaining more positive attitudes towards environmental issues, strengthening ties and improving the ability to teamwork, ultimately translating into more significant learning for both teachers and students in the areas of science, technology and mathematics, among others (Hofstein & Lunetta, 2004; Vannatta, Beyerbach, & Walsh, 2001; Waliczek & Zajicek, 1999).

Promising and innovative, low-cost technologies such as Arduinos have emerged as excellent examples of educational tools to use in hands-on activities and PjBL and to help engage students in STEM related curricula (Banzi, 2009; Banzi & Shiloh, 2014). Arduinos are low-cost hardware and software development boards designed to facilitate the use of electronics and programming (Banzi & Shiloh, 2014).

Each year, the EXPLORA-CONICYT programme finances proposals to develop a one-year project in schools with vulnerability index (VI) greater than 60%. This index uses a mix of socioeconomic variables calculated on the basis of the level of schooling of the mother, father, and household income of each student (Cornejo et al., 2005). The programme represents an opportunity for extremely vulnerable communities in rural sectors to access science and technology projects and thus have educational experiences ex situ.

Our laboratory received funding from this platform in 2015 with the goal of teaching EE and contribute to digital literacy using emerging technologies (e.g. Arduinos) to children from vulnerable schools applying hands-on and PjBL methodologies. With the goal of better understanding the utility of using low-cost emerging technology as a pedagogical resource and the effectiveness of hands-on activities and PjBL, we decided to measure students' motivation in learning science, recording their predisposition and experience during the workshops in which they participated. Our main questions were whether these new, low-cost technologies can be used to change perspectives in learning STEM and in what context we can use these tools to help students being more enthusiastic about STEM subjects. A secondary aim was to relate details of this educational experience and to offer the opportunity to repeat and/or improve it, sharing the design of our low-cost programme for its free use and distribution (Alò et al., 2015).

This communication is a description and evaluation of our experience. We analysed 367 surveys taken during the execution of our project, which allowed us to evaluate: (i) the attitude of underprivileged public-school students towards science and technology; (ii) the predisposition to use new technologies with an experimental approach; and (iii) the perception of underprivileged public-school students towards learning in an informal educational context.

Methods

Educational experiences

The 'Ecoinformática para Jóvenes' (Ecoinformatics for children) initiative took place between March and November 2016 at the Universidad Austral de Chile (UACh) with seventh grade students attending five public schools chosen within the city of Valdivia, Chile with a vulnerability index higher than 60%, according to information provided by the Junta Nacional de Auxilio Escolar y Becas (JUNAEB, 2016) based on the Sistema Nacional de Asignación con Equidad (National Equity Allocation System) (Cornejo et al., 2005). Participating public schools were: Escuela Francia (VI: 90.2%), Instituto Italia (VI: 73.1%), Escuela Alemania (VI: 78.4%), Escuela El Bosque (VI: 79.9%) and Escuela Fernando Santiván (VI: 86.8%). Five workshops were held on the premises of UACh using computer laboratories, classrooms and outdoor spaces (e.g. botanical garden) as needed during the development of the project. A total of 115 students participated in the workshops and were separated into two groups that visited UACh on different days of the week. EE topics addressed climate and global change corresponding to contents selected from the Chilean seventh grade's school curriculum, with applications focused on science, mathematics, and ED-TECH (Table 1). Meanwhile, students organised in groups of three, built a small environmental monitoring station using Arduino technology (Gertz & Di Justo, 2012) (Figure 1). Detailed information about each workshop and instructions for building the environmental monitoring station are available through figshare (Alò et al., 2015). Once the environmental monitor was built following the instructions given, the students had the possibility to use the built-in tool to formulate different types of questions considering the environmental problems presented in each of the workshops. Following the scientific method and with the support of their teachers, each group developed a research project, where they applied their new knowledge about the scientific method using the environmental monitor built with Arduino technology. The final research was presented at two venues: 1. Electronics Fair organised by the student centre of the School of Electronic Engineering of UACh Expotrónica 2016 and 2. Regional Congress of Science and Technology organised by Proyecto Asociativo Regional EXPLORA Los Ríos 2016.

Perception surveys

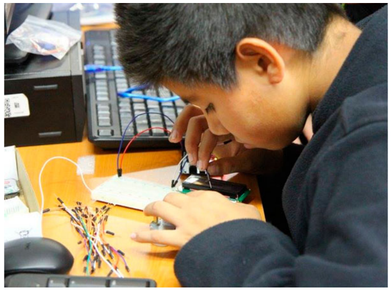
To assess the perception of students towards science and their predisposition in using emerging technologies as tools to facilitate learning, a total of 20 questions were asked before (pre) and after (post) the workshops. The questionnaires were created following previous examples (Clark, Majumdar, Bhattacharjee, & Hanks, 2015; Gautreau & Binns, 2012) and structured with both open and closed-ended questions, where answers were based on a Likert scale (Likert, 1932) with a set of statements for which five levels of response were categorically classified ranging from 1 = 'strongly disagree'; 2 = 'disagree'; 3 = 'neither agree nor disagree'; 4 = 'agree'; 5 = 'strongly agree'. We considered a 'negative attitude' towards the question when the response was close to 1, a 'neutral attitude' above 3, and a 'positive attitude' when the value was close to 5.

The pre-workshop survey considered seven closed-ended questions and three open-ended questions, applied with the objective of obtaining a broader view of their attitude towards the study of science and the methodologies that may favour its learning (Annex 1). The post-workshop survey included 10 closed-ended questions (Annex 2). The pre-workshop survey was applied in the first two workshops of the programme, while the post-workshop survey was applied in workshop 5, with the aim of comparing experiences at the beginning and end of the project. A total 367 surveys were included in the analysis.

Quantitative analysis

Closed-ended questions were analysed using a Student's *t*-test to examine the significance of the average response value of each statement. The test was applied considering a hypothetical sample mean equal to 3 ('neutral') on the Likert scale. A chi-squared test was also applied to assess the existence of statistically significant differences in the attitudes of students before and after participating in the project activities (pre-workshop and post-workshop). We used the *stats* package version 3.2.5 in R studio (R Core Team, 2017), under the null hypothesis of independence of the samples.

Table 1. Description of the curricular contents applied under the hands-on approach in the workshops of 'Ecoinformática para jóvenes'.

Workshop	Science	Technology/Engeneering	Mathematics
Workshop 1 – Global change 	Main causes and consequences of climate change: – Increase in trends of average global temperatures – Greenhouse effects – Exponential increase in human population and urbanisation – Effects on natural resources and the economy – Effects on biodiversity and ecosystems – Carbon and ecological footprints	○ Introduction to electronics and use of Arduino UNO board: operating voltage, analog and digital inputs, Ohm's Law ○ LDC screen connection and programming for text to appear ○ LED light connection and loop programming to turn it on and off in a certain amount of time	Application of Ohm's Law to complete circuits (with LCD display and LED lights)
Workshop 2 – Atmospheric pollution 	– The atmosphere and its composition – Types of atmospheric pollution and possible mitigations – Destruction of the ozone layer and retreat of glaciers	1. Connection and programming of gas sensor MQ135 2. Calibration of sensor to measure CO_2 in ppm	Construction of tables with collected data and comparative graphs of CO2 concentration in different environments

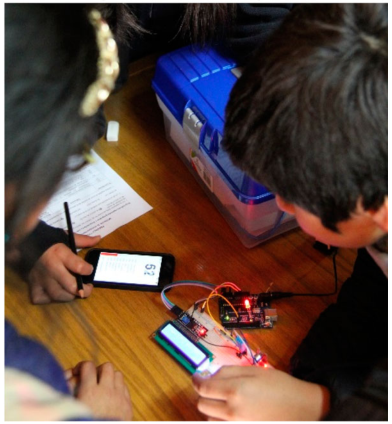
(*Continued*)

Table 1. Continued.

Workshop	Science	Technology/Engeneering	Mathematics
Workshop 3 – Global warming 	– Photosynthesis: how it happens and its importance. – Effects of global warming on vegetation, and hydrological cycles. – Increase in CO_2 concentration.	1. Connection and programming of humidity and temperature sensor DHT11 2. Calibration of the sensor to measure temperature in °C and relative humidity in %.	Temperature and relative humidity graphs in vegetated and unvegetated controlled environments

(*Continued*)

Table 1. Continued.

Workshop	Science	Technology/Engeneering	Mathematics
Workshop 4 – Acoustic pollution 	– Noise pollution and how to measure it – Effects of noise pollution on human health and wildlife – Noise in the oceans – Acoustic pollution in cities (noise map) and mitigation	1. Connection and programming of sound sensor KY038. 2. Sensor calibration to measure noise saturation on a scale from 0% to 100%	Comparative graphs of noise pollution in different environments
Workshop 5 – Renewable energy 	– Types of renewable energies, advantages, disadvantages, and examples in Chile – Benefits of the use of renewable energies – Use of firewood in Chile, effects of using moist firewood and overexploitation of native forests	1. Construction of hygrometer with copper nails and resistance of 1 megaOhm. 2. Programming and calibration to measure moisture content in wood in %	Moisture graphs in firewood samples comparing different tree species (native and introduced)

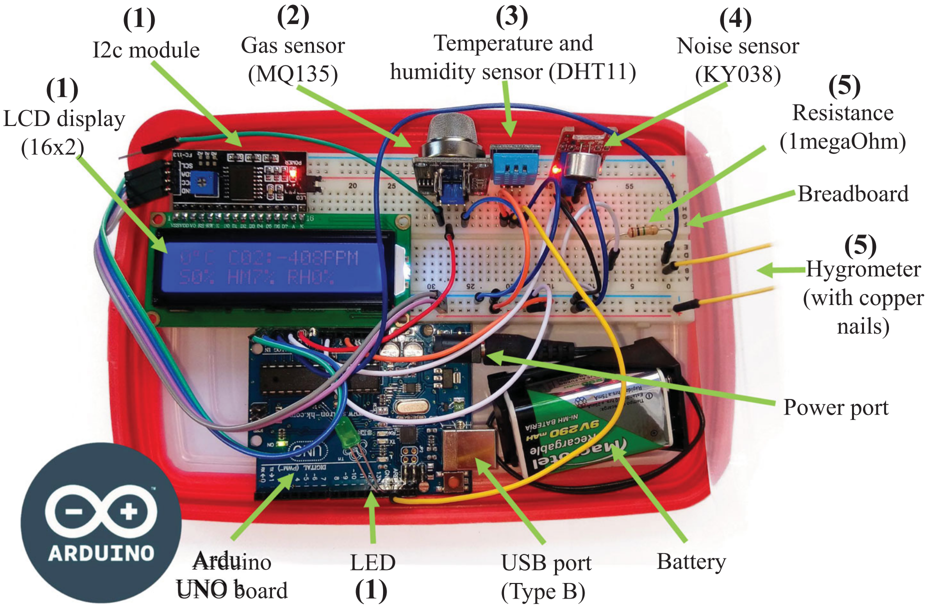


Figure 1. Small environmental monitoring station based on Arduino technology. The numbers indicate in which workshop the connection and programming of the different components and sensors was carried out, as well as the environmental topic that was approached to contextualise the practical work of the students. Where (1) = Workshop 1 - Climate Change; (2) = Workshop 2 - Air Pollution; (3) = Workshop 3 - Global Warming; (4) = Workshop 4 - Acoustic Pollution; and (5) = Workshop 5 - Renewable Energies.

The information collected was separated according to each specific objective and basic descriptive statistics were applied to each in the following order: (I) students' attitude towards science and technology; (II) students' perception using new technologies with a hands-on methodology; (III) learning experience outside the classroom (informal educational context).

Qualitative analysis

Open-ended questions were analysed with the software ATLAS.ti, version 8 (Friese, 2014). ATLAS.ti allows the researcher to associate codes or labels with fragments of text that cannot be analysed significantly with formal and statistical approaches, as well as to search for patterns and classify them (Hwang, 2008; Lewis, 2004). This software allows to identify three levels of information: (i) a 'quote' or segment of text selected by the researcher, which in our case corresponded to a response from the surveyed students; (ii) the 'code' or quote label, which may or not be a concept derived from the same text, and in our case it was a keyword assigned to a concept to analyse; and (iii) group of codes (or 'families' in previous versions of the programme) referring to thematic categories (Fereday & Muir-Cochrane, 2006; Saldaña, 2015).

We imported original transcripts (primary files) from each survey and used the open list coding strategy (Strauss & Corbin, 1990) of inductive reasoning to identify keywords in the text (Gibbs, 2018) and codes emerging from the data (Gallardo, 2014). We assigned names to all the codes; looking to maximise congruence with the concepts they described and minimise the number of codes to analyse. After coding, we followed the approach of Braun and Clarke (2006), applying a thematic analysis to identify specific topics within the available information. These topics were transformed into new groups of codes and represented concepts and opinions of the participants, which allowed comparing answers between different workshops (Alhojailan, 2012). Three groups of codes were defined a priori: (i) how to learn science; (ii) the best of science; (iii) the worst of science.

In order to minimise bias in the results, a single researcher performed all the analyses with ATLAS.ti. Analysis of surveys is a highly iterative process that requires the reading of primary files, coding, revision, re-coding, and finally a more extensive identification and classification into groups or 'families' (Fereday & Muir-Cochrane, 2006; Saldaña, 2015).

Semantic networks

The analysis applied to the open-ended questions allowed us to make connections between the different parts of the information collected and interpret it in more depth. We organised the codes through relationship networks that graphically represented possible structures or systems of relationships between codes and/or groups of codes (Attride-Stirling, 2001). This exercise allowed us to synthesise the main concepts and their connections (Strauss & Corbin, 1990), creating new groups of codes. We worked with networks focused on the groups of codes selected a priori, which represented the initial nodes of the network, and assigned different colours to the codes to represent the groups or 'families' that emerged from this practice. We also established relationships between the codes that co-occur in the different groups, differentiating between group-code relationships and code-code relationships.

Results

Dimension matrix

This research evaluated the attitude of students of underprivileged schools in southern Chile towards the integration of new technologies and hands-on learning activities as tools to address EE and ED TECH curricula. Results for the quantitative analyses have been synthesised in Table 2.

The first dimension (I) evaluated the students' attitude towards science, and, in general, students did not identify with the statement made before or after the workshops ($\text{Mean}_{\text{pre-work}} = 2.9$; $\text{Mean}_{\text{post-work}} = 2.9$). 'I am curious' statement averages a 'more positive' attitude prior to the workshops ($\text{Mean}_{\text{pre-work}} = 3.6$; $\text{SD}_{\text{pre-work}} = 1.3$; t $\text{test}_{\text{pre-work}} = 3.9$; $p < 0.000$), a situation that is sustained after the workshops ($\text{Mean}_{\text{post-work}} = 3.6$; $\text{SD}_{\text{post-work}} = 1.2$; t $\text{test}_{\text{post-work}} = 5.3$; $p < 0.000$). 'My friends see me as a scientist' has a negative connotation prior to the workshops ($\text{Mean}_{\text{pre-work}} = 1.9$; $\text{SD}_{\text{pre-work}} = 1.1$; t $\text{test}_{\text{pre-work}} = -9.5$; $p < 0.000$), which improves slightly after the workshops ($\text{Mean}_{\text{post-work}} = 2.1$; $\text{SD}_{\text{post-work}} = 1.2$; t $\text{test}_{\text{post-work}} = -7.6$; $p < 0.000$). While 'I use what I learn in science classes in my daily life' does not differ statistically from a neutral response, neither in pre-work surveys nor after work in workshops ($\text{Mean}_{\text{pre-work}} = 2.9$; $\text{SD}_{\text{pre-work}} = 1.1$; t $\text{test}_{\text{pre-work}} = -0.9$; $p > 0.05$; and, $\text{Mean}_{\text{post-work}} = 2.9$; $\text{SD}_{\text{post-work}} = 1.2$; t $\text{test}_{\text{post-work}} = -0.7$; $p > 0.05$).

Table 2. Dimensions matrix evaluated with t test on attitudes of teachers and students according to dimensions of analysis, and paired differences between pre-workshop and post-workshop surveys with chi-squared test.

I	Students' attitude towards science and technology
II	Students' attitude using new technologies with a hands-on methodology
III	Learning experience outside the classroom

		Pre-workshops					Post-workshops					Paired Differences	
Dimension	Question	N	Mean	SD	t test	*p* value	N	Mean	SD	t test	*p* value	chi-squared	*p*-value
I	I view myself as good at science	94	3.3	0.9	3.29	1.43E-03	90	3.13	1.08	1.17	2.46E-01	4.99	0.28
	I am curious	94	3.6	1.3	3.98	1.39E-04	90	3.64	1.16	5.25	1.02E-06	0.94	0.96
	My friends see me as a scientist	94	1.9	1.1	−9.49	2.51E-15	90	2.06	1.18	−7.57	3.30E-11	2.04	0.84
	I use what I learn in science classes in my daily life	94	2.9	1.1	−0.90	3.72E-01	90	2.91	1.21	−0.69	4.89E-01	4.75	0.31
II	I learned new things / something new in each of the workshops	100	4.3	1.3	10.39	1.55E-17	83	4.54	0.77	18.25	1.30E-30	4.11	0.39
	I felt comfortable / I found that the atmosphere in which the workshops took place was good	100	4.12	1.1	9.83	2.59E-16	83	4.19	1.04	10.44	1.04E-16	0.5	0.97
	Ecoinformatics and working with Arduinos excites me	100	4.41	0.9	15.11	1.86E-27		4.35	0.94	13.04	1.07E-21	2.35	0.79
	I want to go back to the next workshop series / If it was up to me, I would participate in these workshops again	100	4.68	0.9	18.96	1.02E-34	83	4.47	0.77	17.38	3.15E-29	18.15	0
III	I liked coming to the University / It was an advantage to have done the workshops at the University	100	4.65	0.9	17.63	2.68E-32	83	3.70	1.23	5.19	1.51E-06	6.89	0.22
	My surroundings are a good place to learn science	94	3.6	1.2	5.11	1.72E-06	90	3.42	1.11	3.60	5.16E-04	3.38	0.64
	I see my community as a living laboratory	94	2.6	1.3	−2.75	7.25E-03	90	2.77	1.18	−1.87	6.41E-02	3.08	0.68
	The lab is the best place to learn science	94	4.3	1.0	12.08	9.19E-21	90	3.88	1.15	7.24	1.51E-10	7.56	0.1

Significance levels for *t* test = $p < 0.05$; $p < 0.01$. Significance level for chi-square test = $p > 0.05$; $p < 0.001$.

Results for the second dimension (II) indicate a general positive attitude of the students about use new technologies and hands-on learning. ($Mean_{pre\text{-}work} = 4.4$; $Mean_{post\text{-}work} = 4.3$). The statement, 'I learned new things' shows a more positive attitude after the workshops ($Mean_{pre\text{-}work} = 4.3$; $SD_{pre\text{-}work} = 1.3$; t $test_{pre\text{-}work} = 10.4$; $p < 0.001$; versus, $Mean_{post\text{-}work} = 4.5$; $SD_{post\text{-}work} = 0.8$; t $test_{post\text{-}work} = 18.2$; $p < 0.001$). The same applies to the statement 'I felt comfortable' ($Mean_{pre\text{-}work} = 4.1$; $SD_{pre\text{-}work} = 1.1$; t $test_{pre\text{-}work} = 9.8$; $p < 0.001$; versus, $Mean_{post\text{-}work} = 4.2$; $SD_{post\text{-}work} = 1$; t $test_{post\text{-}work} = 10.4$; $p < 0.001$).

The results of the third dimension (III) on the learning experience outside the classroom, show different attitudes from a positive reaction to the indifference ('neutral') and negativity in some cases. These results are less drastic with the post-workshop surveys ($Mean_{pre\text{-}work} = 3.8$; $Mean_{post\text{-}work} = 3.4$). The statement 'I see my community as a living laboratory' is the one that reports the most negative attitude prior to the workshops ($Mean_{pre\text{-}work} = 2.6$; $SD_{pre\text{-}work} = 1.3$; t $test_{pre\text{-}work} = -2.7$; $p < 0.001$), but the only one that varies favourably towards a more 'neutral' attitude after the workshops ended ($Mean_{post\text{-}work} = 2.8$; $SD_{post\text{-}work} = 1.2$; t $test_{post\text{-}work} = -1.9$; $p > 0.001$)..

The chi-square test shows the existence of statistically significant differences between pre-workshop and post-workshop surveys in each of the statements evaluated in this paper.

Semantic network

Qualitative analysis was applied to 184 student surveys containing open-ended questions, in which 666 citations were identified and classified into 26 codes. Annex 3 details the meaning of each code and the keywords that were used in the process.

A semantic network was constructed based on the classification of the 26 codes into the three groups of codes or 'families' that this paper defined a priori: (i) How to learn science; (ii) The best of science; and (iii) The worst of science (Figure 2). As a result of this exercise, three new families emerged from the data. The first set of codes is associated with the 'educational environment,' in which reference is made to the spaces that are most conducive to science learning according to the students surveyed (i.e. laboratory, outdoor environments, computer room, etc.). Seven codes were grouped under this family (yellow boxes, Figure 2). A second 'family' describes the 'teaching methods', such as the techniques, activities and/or pedagogical resources with which students indicate that they learn science better (i.e. research, experimentation, teacher assistance, etc.). This group consists of 11 codes (blue boxes, Figure 2). Finally, a third group of codes is related to the 'student's attitude' toward science (i.e. uncertainty, new knowledge, socio-environmental problems, among others). In this 'family' we grouped five codes (pink boxes, Figure 2).

Discussion

Chile is going through a deep social crisis where one of the main demands includes improving the education system and to end with the historical segregation driven by education provided for profit, social and economic inequality and burdensome student loan debt (Atria, 2012). In parallel, the scientific and academic community is also demanding a budget increase for science and technology innovation.[2]

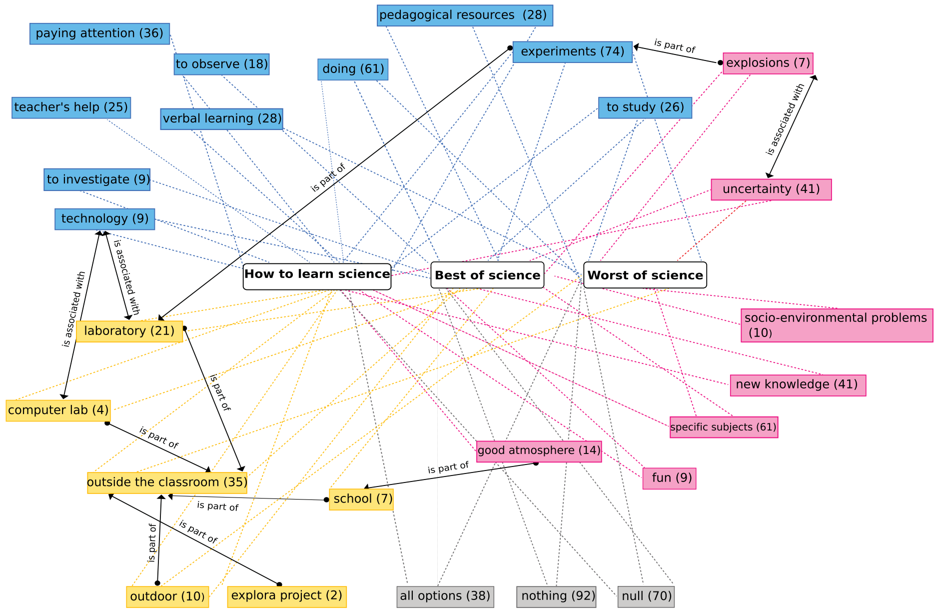


Figure 2. Semantic network for the open-ended questions analysed with ATLAS.ti. Coloured boxes represent the identified codes and their frequency in the analysed surveys. Pink boxes group the family of 'student's attitude', yellow boxes as 'educational environment', and blue boxes as 'teaching method'. White boxes represent the three groups of codes or 'families' defined a priori by the researchers in this study. Dotted lines represent code-family links and continuous lines represent code-code links. Group-code relationships and code-code relationships are represented with one-way and two-way arrows respectively.

The need for Chile to improve and equalise the quality of education to all social strata (Contreras & Elacqua, 2005) requires wise decision-making on the selection of learning contents and its rationale, an issue that has only recently began to be addressed through curricular reform and adjustment processes (Cox, 2011). From a global perspective, current socio-environmental problems and the pace of the technological revolution require well-informed citizens and achievement of core 21 CC (Bayram, 2012; Cox, 2011; Reimers & Chung, 2019). However, although 21 CC have been incorporated into the Chilean curricula according to international guidelines, their implementation in Chilean schools has been inconsistent, possibly due to weaknesses in instructional material, teacher training, and student assessment tools (Bellei & Morawietz, 2016).

Some projects have shown that emerging technologies as hands-on and PjBL didactic tools are an effective alternative in achieving significant learning in a wide range of scientific areas (Badia & García, 2006; Flick, 1993; Thomas, 2000; Tseng, Chang, Lou, & Chen, 2013). The joint application of these tools/strategies can build optimal learning environments to meet the curricular demands of primary and secondary schooling in terms of environmental education (Torres et al., 2017) and technological literacy (González Campos et al., 2017). Likewise, the implementation of methodologies promoting collaborative work for teaching science and technology stimulates educational interaction among students, encourages autonomy, responsibility, self-direction, communication, creativity, and motivation (Acar, 2013; Badia & García, 2006; Wurdinger & Qureshi, 2015) considered

21CC (Bellei & Morawietz, 2016). Hands-on science activities and PjBL have the potential to generate a new educational paradigm where the emphasis is on the student, while the teacher acquires the role of mediator and facilitator, allowing the development of competencies for the resolution of specific problems (González Campos et al., 2017).

Studies that evaluate the implementation of didactic teaching methods similar to those described in this work have demonstrated positive and significant results, independent of age group, educational level or social stratum (González Campos et al., 2017; Gültek, 2005; Tseng et al., 2013). In the case of our study, closed-ended and open-ended surveys provide a view of students' self-concept and attitudes towards science and technology, and their predisposition to integrate different methods and tools for learning these subjects. When we analysed the attitude of students in the three dimensions of interest, we found disparate results. In general, underprivileged children in Chilean public schools manifest indifference and in some cases a negative attitude towards science and technology indicating that some students do not see themselves as scientists, nor do they feel as scientific enquirers (dimension I, table 2). A closer look at the qualitative analysis allows a better understanding of the indifference towards science found among Chilean seventh grade students. The codes grouped under 'student's attitude' (pink boxes, Figure 2) allowed to provide a richer view of the 'uncertainties' student face when confronted to a scientific understanding of nature (e.g '*Science is very difficult*'). Interestingly, this category classified all those answers where fears of the students are made explicit, such as failure (e.g. '*I could make mistakes*'), fear of not being able to understand the curricular contents (e.g '*It is hard for me to learn science*'). According to some authors as Tseng et al. (2013), a negative attitude towards science is justified by the abstract nature and complexity of some subjects (Piburn & Baker, 1993). Traditional teaching methods generally focus on theoretical understanding and memorisation rather than applied work, so students may find science boring and impractical (Mamlok-Naaman, Ben-Zvi, Hofstein, Menis, & Erduran, 2005; Nolen, 2003). Another aspect of the negative attitude towards science emerged from the survey's statement: ' I use what I learn in science classes in my daily life', where students indicated that while technology can be beneficial to everyday life, it may also cause negative effects on society and the environment (e.g. '*Science can be bad for the environment*' in socio-environmental problems code, Figure 2), as also reported by Jenkis (2006) and Tseng et al. (2013). An example about the fear of the potential results stemming from experimenting with science is the sub-category 'explosions', which revealed that at least one group of students related science to chemical experiments, and that these can result in explosions (e.g '*I am afraid that something will explode*'). In general, despite some of the positivity that emerged from the students' scientific self-concept, a neutral attitude towards the survey's statement: 'I am curious' was maintained throughout the workshops. Some students expressed interest in the concepts 'new knowledge' and 'specific subjects' which underscores their interest in learning about environmental science and technology (e.g. '*What I like about science is that I can learn new things and teach them to my family*'; '*What I like about science is that I can learn about animals, plants and nature*'). Engaging activities can provide learning in an entertaining way, enhancing the students' inclination in learning science. Coding of the word 'fun', confirmed this idea from the student's perspective (e.g. '*Having fun I learn much more*'). This concept also emerged in previously studies where the students are not willing to take a science class seriously when they find it to be boring (Gültek, 2005; Mamlok-Naaman et al., 2005).

A positive predisposition of students towards the use of new technologies to learn science emerged in our survey (Dimension II, table 2), (e.g. '*Ecoinformatics and working with Arduinos excites me*'), as well as the students' disposition to continue participating in activities such as our projects. This was supported by the qualitative data analysis. In particular, the group of codes associated to the 'teaching method' (blue boxes, Figure 2) highlights the category 'doing'. This code classified all the opinions of students who indicated that they learn science best through hands-on activities (e.g. '*I learn science better by doing things*'). Within this group of emerging themes, 'experiments' gathered a very high frequency of responses, where students indicated that learning would be more significant if they could see or put into practice what they learned in the laboratory (e.g. '*I learn better doing experiments*'). This shows that students are more inclined to learn science in a practical way as also reported by Tseng et al. (2013). We also found other emerging themes with which students feel closer to understanding science such as 'to investigate' (e.g. '*Doing research projects*'), 'to observe' (e.g. '*I learn science better by observing*'), 'pedagogical resources' (e.g. '*I learn science better with videos and models*' or '*Working in a group*') among others. While students are not necessarily familiar with the 'constructivist philosophy' definition, they mentioned diverse educational strategies based on the hand-on and PjBL methods. The code 'technology' was identified on several occasion (e.g. '*What I like most about science are the Arduinos robots*'), which suggests Arduinos are well received among children and good candidates as educational tools. Students favour working with new technology because it makes STEM subjects more interesting to learn (Jenkis, 2006).

Finally, the assessment of the effectiveness of student learning in environments outside the classroom (Dimension III) has different facets. The students' attitude towards some statements was positive when we asked about their experience at the university ('I liked coming to the University'). In general, learning in informal school environments (e.g. daytrip to the university) was judged as a positive event by students and widened the collective perception of the laboratory as the only space to develop science. For example, when the survey stated: 'The lab is the best place to learn science', the students' attitude changed from positive to neutral, possibly because during the development of our project the students realised that 'science' can also be performed in outdoor environments. Our experiments were not exclusive to the classroom or laboratories. While the student's attitude on the statement: 'My surroundings are a good place to learn science' was always neutral throughout the workshops. Following this idea, occasional indifference was probably associated with statements that sought to broaden the view of opportunities to explore science in environments beyond the classroom. The students' perception about places apt for science was reflected in the group of codes 'educational environment' (yellow boxes, Figure 2), which classified concepts as 'laboratory' (e.g. '*I learn science better in the laboratory*'), and links the codes 'experiments' and 'technology' (blue boxes, Figure 2), and 'computer room' (e.g. '*I learn science better in the computer room*') There is also the concept of 'open air' (e.g. '*Investigate unusual things in open air*') which represents ideas as varied as 'schoolyard' and 'field trips'. This last concept is interesting because it reflects the student's demand to learn science in natural environments. Previous studies such as Waite (2011) showed children's love for the outdoor through their demeanour and testimony about their learning experience.

This project sought to apply hands-on and PjBL methodologies outside the classroom, generating a new learning experience for the students. Probably because of its short duration, our project did not generate a significant change in students' attitude towards science, however we noted a positive disposition and interest in science's subjects as well as a desire to do experiments and hands-on work. In this sense, the experience of developing a small environmental monitoring station with Arduino technology as a PjBL activity gave the students an opportunity to learn through group effort, discussion and continuous examination applying the scientific method. Indeed, hands-on activities are necessary to generate concrete experiences, especially in students who have had little exposure to significant learning about the natural environment (Flick, 1993).

Conclusions

At present, scarce pedagogical material and training are provided to teachers and it is not clear how to implement the environmental contents in the framework of the educational reform given the low political priorities assigned to EE in Chile (Torres et al., 2017). Our project showed the attitudes of a group of students from Chilean public schools toward science and exposing possible limitations will contribute to improve its future implementation in other instances. For example, due to the transversal nature of EE teaching in Chile the willingness to implement similar programmes is often limited to the teachers' initiative, which is ultimately determined by the teacher's curiosity, enthusiasm and content knowledge. Moreover, Chilean teachers are often compelled to extensive work schedules, with a limited amount of time left for professional development or to participate in extra-curricular activities (Zorrilla, 2016; Torres et al., 2017). Additionally, even though the experience presented here does not require large resources to be replicated (Arduino technology has been specifically designed to teach and learn electronics at low-costs and with a high variety of applications; Arduino, 2018; Banzi, 2009; Banzi & Shiloh, 2014; Gertz & Di Justo, 2012), it requires the use of personal computers to programme the Arduinos and the purchase of additional hardware (e.g. Arduinos, sensors etc.). However, while most Chilean public schools are now equipped with computer labs, schools with a high index of vulnerability generally have fewer possibilities to have access to this type of resources/activities. Therefore, hardware's costs, even if relatively low, could still be burdensome due to the budget restrictions imposed on the Chilean public-school system.[3]

To replicate our project or to develop similar ones, we recommend the active participation of the school STEM teachers along the different facets of the interdisciplinary learning process (e.g. programming and assembling Arduinos assisted by ED TECH teachers; data collection and analysis assisted by math teachers; environmental problems applications assisted by natural science teachers). To start teaching programming skills to children, visual programming languages (VPL) are a good alternative to traditional text-based coding because they use animations (Booth & Stumpf, 2013). The VPLs implements a modified Scratch block language (Resnick et al., 2009) and provides commands' syntax already written (or 'blocks') to control sensors and actuators connected to the Arduino board. Beginning programmers can choose these functions from a palette of available blocks, schematically visualise them in the interface and just keep focus on code building and not on the syntaxis (Booth & Stumpf, 2013). There are several alternatives such as S4A (Scratch for Arduino, http://seaside.citilab.eu/scratch/arduino), Minibloq (http://blog.minibloq.org/), ArduBlock (http://blog.ardublock.com/) and Modkit (http://www.modk.it/). Another option is Lego Mindstorms (http://mindstorms.lego.com/), a line of hardware that teaches robotics and programming in a variety of languages as JAVA and C++. However, this technology has non-open robotics kits with a proprietary programming language, which makes it more expensive that Arduino (López-Rodríguez & Cuesta, 2016). These new teaching and learning tools will likely have a greater and long-lasting educational impact on the students, promoting faster learning, problem solving ability and foster their critical thinking skills, among other benefits (Clark et al., 2015; Gautreau & Binns, 2012; Tseng et al., 2013).

Given that the students who participated in our workshops highly valued developing the project outside the classroom, we recommend taking activities outdoors or using spaces such as computer laboratories. In particular, the out-of-school experience for students participating in our workshops was based at the local university premises which, for low-income students[4], could also mean increasing their motivation to pursue a higher education degree. Furthermore, the development of positive attitudes towards science and technology can motivate young people's interest in science education and science-related careers (Crawley III & Coe, 1990; Norwich & Duncan, 1990).

Constructivist philosophy encourages learning outside the classroom, and children begin life as exploratory learners enjoying the rich experimental qualities of informal contexts (Beames, Higgins, & Nicol, 2012; Waite, 2011). Therefore, the inclusion of methodologies such as hands-on activities or PjBL can increase students' learning interest and further facilitate the development of both cognitive and social 21CC skills (Flick, 1993; Tseng et al., 2013) and to cope with a rapidly changing society and environment. Our research provides new background in the application of constructivist methodologies and use of emerging technologies in a developing country and raises further questions on which is the best approach to teach and learn environmental science and technological literacy in primary and secondary schools.

Notes

1. SIMCE - Sistema de Medición de la Calidad de la Educación
2. Investment in research and development was only 0.38% of gross domestic product (GDP) in 2010 (Ministerio de Economía del Gobierno de Chile, 2012), a very low value compared to the OCDE average of 2.5% of GDP.
3. At 3.4% of the gross domestic product, spending on primary, secondary and post-secondary educational institutions in Chile was still below OECD average in 2013 (OECD, 2017).
4. Income inequality is associated with lower education attainment and Chile has the greatest inequalities on educational attainment in the OECD (OECD, 2017).

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

Acar, G. (2013). The effect of project-based learning on students' motivation. International Journal of Academic Research, 5(2), 82–86.

Alhojailan, M. I. (2012). Thematic analysis: A critical review of its process and evaluation. West East Journal of Social Sciences, 1(1), 39–47.

Alò, D., Castillo, A.G., Rojas, R., Gho-Illanes, D., Rodriguez, C., & Samaniego, H. (2015). Ecoinformatica para Jovenes [Ecoinformatics for children: capturing environmental information to understand our environment]. *Universidad Austral de Chile*. https://doi.org/10.6084/m9.figshare.9597272.v1

Arduino. (2018). Arduino - Introduction. Retrieved from https://www.arduino.cc/

Atria, F. (2012). *La mala educación: Ideas que inspiran al movimiento estudiantil de Chile [Bad education: Ideas that inspire Chile's student movement]*. Editorial Catalonia.

Attride-Stirling, J. (2001). Thematic networks: An analytic tool for qualitative research. *Qualitative Research*, 1(3), 385–405.

Badia, A., & García, C. (2006). Incorporación de las TIC en la enseñanza y el aprendizaje basados en la elaboración colaborativa de proyectos [Incorporation of ITCs in teaching and learning based on collaborative project development]. *Revista de Universidad y Sociedad Del Conocimiento*, 3(2), 42–54.

Banzi, M. (2009). *Getting started with Arduino*. Sebastopol: O'Reilly Media, Inc.

Banzi, M., & Shiloh, M. (2014). Getting started with Arduino: the open source electronics prototyping platform. Maker Media, Inc.

Bayram, S. A. (2012). On the role of Intrinsic value in terms of environmental education. *Procedia - Social and Behavioral Sciences*, 47, 1087–1091.

Beames, S., Higgins, P., & Nicol, R. (2012). *Learning outside the classroom theory and guidelines for practice*. New York and London: Routledge. https://doi.org/10.1073/pnas.1908871116

Bellei, C. (2013). El estudio de la segregación socioeconómica y académica de la educación chilena [The study of the socioeconomic and academic segregation of Chilean education]. *Estudios Pedagógicos (Valdivia)*, *39*(1), 325–345.

Bellei, C., & Morawietz, L. (2016). Strong context, weak tools: Twenty-first -century-competencies in the chilean educational reform. In F. Reimers & C. K. Chung (Eds.), *Teaching and learning for the twenty-first century. Educational goals, policies and curricula from six nations* (pp. 93–126). Cambridge, MA: Harvard Education Press.

Biblioteca del Congreso Nacional. (2015). Ley Número 20.845 de Inclusión Escolar, Ministerio de Educación. Chile. https://doi.org/10.1016/j.snb.2013.10.100

Booth, T., & Stumpf, S. (2013). End-User experiences of visual and Textual programming environments for Arduino. In Y. Dittrich, M. Burnett, A. Mørch, & D. Redmiles (Eds.), *End-user development. IS-EUD 2013. Lecture notes in computer science* (Vol. 7897, pp. 25–39). Berlin: Springer.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101.

Cabezas, M. C. (1997). *Educación ambiental y lenguaje ecológico. Una propuesta didáctica para la enseñanza de la educación ambiental* [*Environmental education and ecological language. A didactic proposal for the teaching of environmental education*]. Castilla Ediciones.

Clark, L., Majumdar, S., Bhattacharjee, J., & Hanks, A. C. (2015). Creating an atmosphere for STEM literacy in the rural South through student-collected Weather data. *Journal of Geoscience Education*, *63*(2), 105–115.

Contreras, D., & Elacqua, G. (2005). El desafío de la calidad y equidad en la educación chilena [The challenge of quality and equity in Chilean education]. *En Foco-Expansiva*, *30*(43), 1–45.

Cornejo, A., Céspedes, P., Escobar, D., Núñez, R., Reyes, G., & Rojas, K. (2005). SINAE. Sistema Nacional de Asignación con Equidad Para Becas JUNAEB: Una nueva visión en la construcción de igualdad de oportunidades en la infancia [National Equity Allocation System for JUNAEB Scholarships: A new vision in the construction of equal opportunities in childhood]. *Santiago de Chile: Junta Nacional de Auxilio Escolar y Becas, Dirección Nacional.*

Cox, C. (2011). Currículo escolar de Chile: Génesis, implementación y desarrollo [Chile's school curriculum: Genesis, implementation and development]. *Revue International de Education de Sevres*, *56*, 1–9.

Crawley III, F. E., & Coe, A. S. (1990). Determinants of middle school students' intention to enroll in a high school science course: An application of the theory of reasoned action. *Journal of Research in Science Teaching*, *27*(5), 461–476.

Drago, J. L., & Paredes, R. D. (2011). La brecha de la calidad en la educación chilena [The quality gap in Chilean education].

Fereday, J., & Muir-Cochrane, E. (2006). Demonstrating rigor using thematic analysis: A hybrid approach of inductive and deductive coding and theme development. *International Journal of Qualitative Methods*, *5*(1), 80–92.

Flick, L. B. (1993). The meanings of hands-on science. *Journal of Science Teacher Education: The Official Journal of the Association for the Education of Teachers in Science*, *4*(1), 1–8.

Friese, S. (2014). *Qualitative data analysis with ATLAS.ti*. London: SAGE Publications Ltd.

Gallardo, E. (2014). Utilización del programa de análisis cualitativo ATLAS.ti para gestionar y analizar datos [Using the qualitative analysis program ATLAS.ti to manage and analyze data]. Retrieved from https://atlasti.com/2014/06/12/utilizacion-del-programa-de-analisis-cualitativo-atlas-ti-para-gestionar-y-analizar-datos/

Gautreau, B. T., & Binns, I. C. (2012). Investigating student attitudes and achievements in an environmental place-based inquiry in secondary classrooms. *International Journal of Environmental and Science Education*, *7*(2), 167–195.

Gertz, E., & Di Justo, P. (2012). *Environmental monitoring with Arduino: Building simple devices to collect data about the world around us*. Sebastopol, CA: O'Reilly Media, Inc.
Gibbs, G. R. (2018). *Analyzing qualitative data* (Vol. 6). London: Sage.
González Campos, D., Olarte Dussán, F., & Corredor Aristizabal, J. (2017). La alfabetización tecnológica: De la informática al desarrollo de competencias tecnológicas [Technological literacy: From information technology to the development of technological skills]. *Estudios Pedagógicos (Valdivia)*, *43*(1), 193–212.
Gültek, M. (2005). The effect of project based learning on learning outcomes in the 5 th grade social studies course in primary education. *Educational Sciences: Theory & Practice*, 5(November), 548–557.
Hofstein, A., & Lunetta, V. N. (2004). The laboratory in science education: Foundations for the twenty-first century. *Science Education*, *88*(1), 28–54.
Hwang, S. (2008). Utilizing qualitative data analysis software: A review of ATLAS. ti. *Social Science Computer Review*, *26*(4), 519–527.
Jenkis, EW. (2006). Student opinion in England about science and technology. Research in Science & Technological Education, 24(1), 59–68.
Jones, A., Buntting, C., & De Vries, M. J. (2013). The developing field of technology education: A review to look forward. *International Journal of Technology and Design Education*, *23*(2), 191–212.
JUNAEB. (2016). IVE - JUNAEB. Retrieved from https://www.junaeb.cl/ive
Krajcik, J., Czerniak, C., & Berger, C. (1999). *Teaching Children Science: A Project-Based Approach*. edited by M.-H. College. Boston.
Lewis, R. B. (2004). NVivo 2.0 and ATLAS.ti 5.0: A comparative review of two popular qualitative data-analysis programs. *Field Methods*, *16*(4), 439–464.
Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*.
López-Rodríguez, FM., & Cuesta, F. (2016). Andruino-a1: Low-cost educational mobile robot based on android and arduino. *Journal of Intelligent & Robotic Systems*, *81*(1), 63–76.
Mamlok-Naaman, R., Ben-Zvi, R., Hofstein, A., Menis, J., & Erduran, S. (2005). Learning science through a historical approach: Does it affect the attitudes of non-science-oriented students towards science? *International Journal of Science and Mathematics Education*, 3(3), 485–507.
MINEDUC. (2017). *¿Hacia dónde avanza el sistema educativo en Chile? Análisis de las recomendaciones OCDE contenidas en Evaluaciones de Políticas Nacionales de Educación: Educación en Chile (2014-2016) en el contexto de la reforma en marcha [What is the direction of the education system in Chile? Analysis of OCDE recommendations contained in National Education Policy Evaluations: Education in Chile (2014-2016) in the context of the ongoing reform]. Evidencias*. ISBN 978-956-363-274-3
MINEDUC. (2018). Resultados Educativos 2017 [Educational Outcomes 2017] - Agencia de la Calidad de la Educación (pp. 1–49).
Ministerio de Economía del Gobierno de Chile. (2012). Retrieved from (Julio, 2012).
Nolen, S. B. (2003). Learning environment, motivation, and achievement in high school science. *Journal of Research in Science Teaching: The Official Journal of the National Association for Research in Science Teaching*, *40*(4), 347–368.
Norwich, B., & Duncan, J. (1990). Attitudes, subjective norm, perceived preventive factors, intentions and learning science: Testing a modified theory of reasoned action. *British Journal of Educational Psychology*, *60*(3), 312–321.
OCDE. (2018). Estudios Económicos de la OCDE [OECD Economic studies]. *Estudios Económicos de La OCDE*, *1*, 72. Retrieved from https://www.oecd.org/eco/surveys/Chile-2018-OECD-economic-sruvey-Spanish.pdf
OECD. (2017). Education in Chile. In *Reviews of national policies for education*. Paris: OECD Publishing.
Piburn, M. D., & Baker, D. R. (1993). If I were the teacher … qualitative study of attitude toward science. *Science Education*, *77*(4), 393–406.

Programa de las Naciones Unidas para el Medio Ambiente. (1975). La Carta de Belgrado - Un marco general para la educación ambiental [A general framework for environmental education]. In *International workshop on environmental education* (pp. 1–5). Belgrade: Unesco.

R Core Team. (2017). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. Retrieved from https://www.r-project.org/

Reimers, F. M., & Chung, C. K. (Eds.). (2019). *Teaching and learning for the twenty-first century: Educational goals, policies, and curricula from six nations*. Harvard Education Press.

Resnick, M., Maloney, J., Monroy, Hernández, A., Rusk, N., Eastmond, E., Brennan, K., Millner, A., Rosenbaum, E., Silver, J., Silverman, B., & Kafai, Y. (2009). Scratch: Programming for all. Communications of the ACM 52(11), 60-67.

Saldaña, J. (2015). *The coding manual for qualitative researchers*. London: Sage Publications Ltd.

Strauss, A., & Corbin, J. M. (1990). *Basics of qualitative research: Grounded theory procedures and techniques*. London: Sage Publications, Inc.

Thomas, J. W. (2000). *A review of research on project-based learning*. San Rafael, CA: Autodesk Foundation. Retrieved from http://www.autodesk.com/foundation

Torres, L. B., Benavides, J. E., Latoja, C. J., & Novoa, E. R. (2017). Presencia de una Educación Ambiental basada en conocimiento, actitudes y prácticas en la enseñanza de las ciencias naturales en establecimientos municipales de la ciudad de Los Ángeles, Chile [Environmental education based on knowledge, attitudes and practices in natural science teachings in public establishments of the city of Los Angeles, Chile]. *Estudios Pedagógicos*, *43*(3), 311–323.

Tseng, K. H., Chang, C. C., Lou, S. J., & Chen, W. P. (2013). Attitudes towards science, technology, engineering and mathematics (STEM) in a project-based learning (PjBL) environment. *International Journal of Technology and Design Education*, *23*(1), 87–102.

Valenzuela, J. P. (2008). Segregación en el Sistema Escolar Chileno: en la Búsqueda de una educación de calidad en un contexto deextrema desigualdad [Segregation in the Chilean school system: in the search for quality in a context of extreme inequality]. In *Transformaciones del Espacio Público, II Escuela Chile-Francia* (pp. 131–156). Santiago de Chile: Facultad de Arquitectura y Urbanismo, Universidad de Chile.

Valenzuela, J. P., Bellei, C., & De Los Ríos, D. (2010). Segregación escolar en Chile [School segregation in Chile]. In S. Martinic & G. Elacqua (Eds.), *Cambios en la gobernanza del sistema educativo chileno* (pp. 257–284). Santiago de Chile: UNESCO, Pontificia Universidad Católica de Chile.

Valenzuela, J. M., Labarrera, P., & Rodríguez, P. (2008). Educación en Chile-entre la continuidad y las rupturas. Principales hitos de las políticas educativas [education in Chile - between continuity and ruptures. Main milestones of education policies]. *Revista Iberoamericana de Educación*, *48*, 129–145.

Vannatta, R., Beyerbach, B., & Walsh, C. (2001). From teaching technology to using technology to enhance student learning : Preservice teachers' changing perceptions of technology infusion. *Journal of Technology and Teacher Education*, *9*(1), 105–127.

Waite, S. (2011). Teaching and learning outside the classroom: Personal values, alternative pedagogies and standards. *Education 3-13*, *39*(1), 65–82. https://doi.org/10.1080/03004270903206141

Waliczek, T. M., & Zajicek, J. M. (1999). School gardening: Improving environmental attitudes of children through hands-on learning 1. *J. Environ. Hort*, *17*(4), 174–180.

Wurdinger, S., & Qureshi, M. (2015). Enhancing college students' life skills through project based learning. *Innovative Higher Education*, *40*(3), 279–286.

Zorrilla, JP. (2016). La regulación de la jornada docente en perspectiva comparada: Los casos de Argentina, Brasil, Chile y Uruguay [The regulation of the teaching day in a comparative perspective: The cases of Argentina, Brazil, Chile and Uruguay]. *Propuesta Educativa*, 45, 10–20.